QUICK START & SAFETY GUIDE · SKU PIA24024P

From the Box to Live EEG over Bluetooth

Seven steps: flash the NUCLEO-WB55 once, mount the shield, connect the electrodes, power from a battery, connect over Bluetooth from Python, Android, the browser or PiEEG Server, and check the signal.



WARNING Battery only.

MicroBCI has **no isolation circuitry**. Connect the Nucleo to a computer only to flash it, with no electrodes on anyone. While recording it runs from a **small 5 V power bank**. MicroBCI is **not a medical device**.

01 SAFETY RULES — READ BEFORE PUTTING ELECTRODES ON ANYONE

#	Rule
1	**Battery only.** While electrodes are on a person, the Nucleo and MicroBCI run from a small 5 V power bank.
2	**Flash off the person.** Connect the Nucleo to a computer only to flash the firmware. Unplug it before recording.
3	**No mains connection of any kind:** no wall charger, no desk supply, no USB cable to a computer while recording.
4	**No diagnosis.** Do not use MicroBCI to diagnose, monitor or treat any medical condition.
5	**Implanted devices.** Not on a person with a pacemaker or other implant without a physician's approval.
6	**Skin.** Clean, intact skin only. Stop if it becomes irritated; clean off gel after each session.
7	**Exposed electronics.** No case: handle by the edges, use anti-static precautions, keep liquids away.

Power while recording

Setup	While electrodes are on	Why
Nucleo + MicroBCI on a small 5 V power bank	SAFE	No path to the mains.
Computer or phone receiving over Bluetooth	SAFE	No cable to the board or the person.
Nucleo plugged into a computer's USB	NEVER	Links the person to the mains through the computer.
Nucleo on a wall charger	NEVER	Connects the person to the mains through the electrodes.

For research and education only. Not a medical device. Specifications subject to change.

02 WHAT YOU NEED

Item	Requirement
MicroBCI	Included
NUCLEO-WB55	STM32WB55RG board; its built-in ST-LINK is used for flashing
Micro-USB data cable	Flashing from a computer, then power from the bank
5 V power bank	Small: ≤ 1 A, ≤ 1,000 mAh, charged
Electrodes	8 dry or wet Ag/AgCl with 2.54 mm connectors + 2 ear clips (or an EMG / ECG kit)
EEG cap	10-20 cap
Computer	STM32CubeProgrammer (once) and Python 3 with Bluetooth LE, or Chrome / Edge
Android phone	Optional, for the app

03 SETUP IN SEVEN STEPS

1

Gather the parts

MicroBCI, NUCLEO-WB55, Micro-USB data cable, charged power bank, 8 electrodes and 2 ear clips, cap.

2

Flash the Nucleo (first time only)

With STM32CubeProgrammer over the Nucleo's ST-LINK USB port: install `stm32wb5x_BLE_Stack_full_fw.bin` at **0x080CE000**, then download `Micro_BCI.hex`. Full steps in MB-SW-006.

3

Mount the shield

Unplug the Nucleo. Line the shield up with the end pin marked by the arrow on the alignment photo and press down evenly on both morpho headers.

4

Connect the electrodes

Channels 1–8 to the cap, **Ref** to one earlobe, **Biasout** to the other (MB-PIN-004).

5

Power from the battery

Plug the **power bank** into the Nucleo's Micro-USB port. LED D4 lights and the board advertises over Bluetooth as **EAREEG**.

6

Connect over Bluetooth

No pairing needed. Pick one:

- **Python**: set your board's address in SDK/GUI/1.1_GUI.py and run it.
- **Browser**: cloud.pieeg.com in Chrome or Edge → PiEEG Devices → IronBCI.
- **Android**: install mobile_sdk_Micro_BCI.apk, set 8 ch, connect.
- **PiEEG Server**: `pip install "pieeg-server[ironbci]"`, `pieeg-server --device ironbci8`.

7

Check the signal

Sit still, then close the eyes for 10 seconds: alpha (8–12 Hz) rises, then drops when the eyes open.

04 FIRST SIGNAL CHECKS

Test	What you should see
Sit still for 30 s	All 8 channels settle. Flat lines or huge values mean a loose electrode.
Close the eyes for 10 s	Alpha (8–12 Hz) rises, then drops when the eyes open.
Blink (forehead electrodes)	Large slow spikes.
Chew or clench	Bursts of high-frequency noise — muscle (EMG), not brain activity.

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05 TROUBLESHOOTING

Flashing

Problem	Fix
ST-LINK not recognised	Use the Nucleo's ST-LINK USB port and a data cable; install or update the ST-LINK driver.
Cannot connect in CubeProgrammer	Port SWD, programmer ST-LINK, then Connect; unplug and replug the board.
FUS not v1.2.0.0 / v1.3.0.0	Update the FUS with ST's file for your STM32WB first, then install the stack.
Stack upgrade fails	File stm32wb5x_BLE_Stack_full_fw.bin at exactly 0x080CE000.
No advertising after flashing	Flash both the stack and Micro_BCI.hex, then power-cycle.

Bluetooth

Problem	Fix
Python GUI cannot connect	The GUI uses a fixed address: set address to your board's (MB-SW-006).
Device not found	Board powered, stack and firmware installed; move closer.
macOS address format	bleak shows a UUID instead of a MAC address — use the UUID.
Found but will not connect	Already connected elsewhere: close the other app, power-cycle, retry.
No Bluetooth in the browser	Use Chrome or Edge; Safari, Firefox and iOS browsers lack Web Bluetooth.

Data and signal

Problem	Fix
No data, or all values constant	The shield may be shifted by one pin: re-seat and check the alignment.
All channels noisy at once	Check the **REF and BIAS ear clips** first.
One channel flat or very large	That electrode is not touching the skin.
Strong 50/60 Hz hum	Run from the power bank, not a computer's USB; move away from chargers.
Values in the thousands of µV	Normal DC offset: use 1.1_GUI.py (1–40 Hz) or filter the data.

06 RELATED DOCUMENTS

MB-DS-001 Product Datasheet	**MB-MD-002** Mechanical & Dimensional Drawing	**MB-PCB-003** PCB & Manufacturing Specification
MB-PIN-004 Connector Pinout & Electrode Guide	**MB-SW-006** Firmware, Bluetooth & Software Guide	

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